

CHAPTER – 1

INTRODUCTION

1. INTRODUCTION

Agriculture has been the backbone of human civilization for centuries, and it continues to play a vital role in sustaining global economies and ensuring food security. Among various crops cultivated worldwide, bananas hold a significant place due to their high nutritional value, economic importance, and vast consumption across different regions. However, banana cultivation is frequently challenged by numerous diseases that can drastically reduce crop yield and quality. Detecting and managing these diseases at an early stage is crucial for minimizing losses and promoting sustainable farming practices.

In recent years, technological advancements, particularly in Artificial Intelligence (AI) and Machine Learning (ML), have opened new avenues for solving age-old agricultural problems. Image-based disease detection powered by deep learning has emerged as a promising solution that offers rapid, reliable, and scalable methods for identifying plant diseases. This project aims to design and implement an AI-Driven Crop Disease Prediction and Management System focused primarily on banana plants. By leveraging Convolutional Neural Networks (CNNs) and pre-trained deep learning models such as MobileNetV2, EfficientNet, DenseNet, and InceptionV3, the system strives to achieve high-accuracy disease identification from leaf images.

The project not only aims to assist farmers in making informed decisions but also seeks to demonstrate the transformative potential of AI in agriculture. With early detection, farmers can apply targeted treatments.